

A Technical Introduction to Adversarial examples

Section 1

With this boundary function, the attack then follows an iterative algorithm to find adversarial examples x' for a given image x that satisfies the attack objectives.

Section 2

Data poisoning Poisoning consists of contaminating the training dataset with data designed to increase errors in the output. Given that learning algorithms are shaped by their training datasets, poisoning can effectively reprogram algorithms with potentially malicious intent. Concerns have been raised especially for user-generated training data, e.g. for content recommendation or natural language models. The ubiquity of fake accounts offers many opportunities for poisoning. Facebook reportedly removes around 7 billion fake accounts per year. Poisoning has been reported as the leading concern for industrial applications. On social medias, disinformation campaigns attempt to bias recommendation and moderation algorithms, to push certain content over others. A particular case of data poisoning is the backdoor attack, which aims to teach a specific behavior for inputs with a given trigger, e.g. a small defect on images, sounds, videos or texts.

Section 3

Labeling and ground truth challenges Malware labels are often unstable because different antivirus engines may classify the same sample in conflicting ways. Ceschin et al. note that families may be renamed or reorganized over time, causing further discrepancies in ground truth and reducing the reliability of benchmarks.

Section 4

Model extraction Model extraction involves an adversary probing a black box machine learning system in order to extract the data it was trained on. This can cause issues when either the training data or the model itself is sensitive and confidential. For example, model extraction could be used to extract a proprietary stock trading model which the adversary could then use for their own financial benefit. In the extreme case, model extraction can lead to model stealing, which corresponds to extracting a sufficient amount of data from the model to enable the complete reconstruction of the model. On the other hand, membership inference is a targeted model extraction attack, which infers the owner of a data point, often by leveraging the overfitting resulting from poor machine learning practices. Concerningly, this is sometimes achievable even without knowledge or access to a target model's parameters, raising security concerns for models

trained on sensitive data, including but not limited to medical records and/or personally identifiable information. With the emergence of transfer learning and public accessibility of many state of the art machine learning models, tech companies are increasingly drawn to create models based on public ones, giving attackers freely accessible information to the structure and type of model being used.

Section 5

Adversarial attacks and training in linear models There is a growing literature about adversarial attacks in linear models. Indeed, since the seminal work from Goodfellow et al. studying these models in linear models has been an important tool to understand how adversarial attacks affect machine learning models. The analysis of these models is simplified because the computation of adversarial attacks can be simplified in linear regression and classification problems. Moreover, adversarial training is convex in this case. Linear models allow for analytical analysis while still reproducing phenomena observed in state-of-the-art models. One prime example of that is how this model can be used to explain the trade-off between robustness and accuracy. Diverse work indeed provides analysis of adversarial attacks in linear models, including asymptotic analysis for classification and for linear regression. And, finite-sample analysis based on Rademacher complexity. A result from studying adversarial attacks in linear models is that it closely relates to regularization. Under certain conditions, it has been shown that

Section 6

One important property of this equation is that the gradient is calculated with respect to the input image since the goal is to generate an image that maximizes the loss for the original image of true label y . In traditional gradient descent (for model training), the gradient is used to update the weights of the model since the goal is to minimize the loss for the model on a ground truth dataset. The Fast Gradient Sign Method was proposed as a fast way to generate adversarial examples to evade the model, based on the hypothesis that neural networks cannot resist even linear amounts of perturbation to the input. FGSM has shown to be effective in adversarial attacks for image classification and skeletal action recognition.

Section 7

$$\min (||\delta||_p) \text{ subject to } C(x + \delta) = t, x + \delta \in [0, 1]^n$$

Section 8

Performance evaluation The choice of the evaluation metrics can impact the validity of the results. Having an inappropriate baseline involves failing to compare a new

model against simpler, well-established baselines. Inappropriate performance measures means using metrics that do not align with the practical goals of the system. Reporting only "accuracy" is often described as insufficient for an intrusion detection system, where false-positive rates are considered critically important. Base rate fallacy is a failure to correctly interpret performance in the context of large class imbalances.

Section 9

Untargeted: $\min_{x'} d(x', x) \text{ subject to } C(x') \neq C(x)$
$$\{\text{displaystyle } \textbf{Untargeted:} \} \min_{x'} d(x', x) \text{ subject to } C(x') \neq C(x)$$

Section 10

The result in theory is an adversarial example that is highly confident in the incorrect class but is also very similar to the original image. To find such example, Square Attack utilizes the iterative random search technique to randomly perturb the image in hopes of improving the objective function. In each step, the algorithm perturbs only a small square section of pixels, hence the name Square Attack, which terminates as soon as an adversarial example is found in order to improve query efficiency. Finally, since the attack algorithm uses scores and not gradient information, the authors of the paper indicate that this approach is not affected by gradient masking, a common technique formerly used to prevent evasion attacks.

Section 11

System design and learning Feature engineering and training can causes issues. Data snooping is a common pitfall where a model is trained using information that would not be available in a real-world scenario. Spurious correlations result when a model learns to associate artifacts with a label, rather than the underlying security-relevant pattern. For example, a malware classifier might learn to identify a specific compiler artifact instead of malicious behavior itself. Biased parameter selection is a form of data snooping where model hyperparameters are tuned using the test set.